The missing signal

Telling similar entities apart as knowledge changes

Pengyu Zhang

PhD candidate in Computer Science, University of Amsterdam · INDElab
Defense planned for March 2027
p.y.zhang3@gmail.com · pengyu-zhang.github.io



Entity ambiguity meets temporal drift

Entity ambiguity



≈

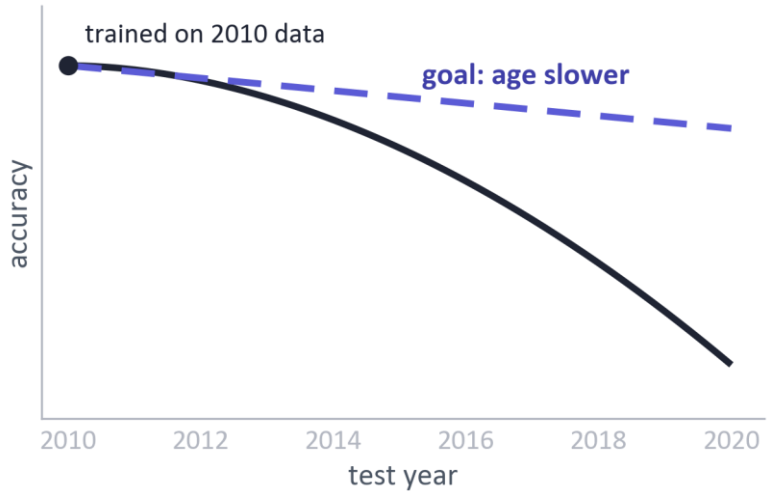


Napoleon Bonaparte
French Revolution
French Republic
Marie Antoinette
Napoleonic Wars

≈

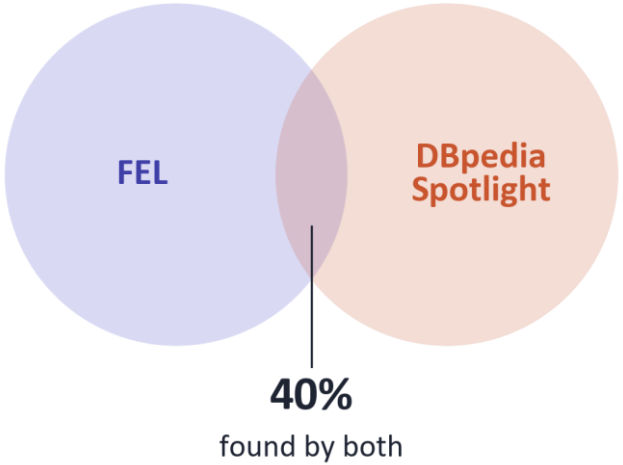
Napoleon (2023 film)
French Revolution
French Republic
Marie Antoinette
French Empire

Temporal drift



Both show up in real data

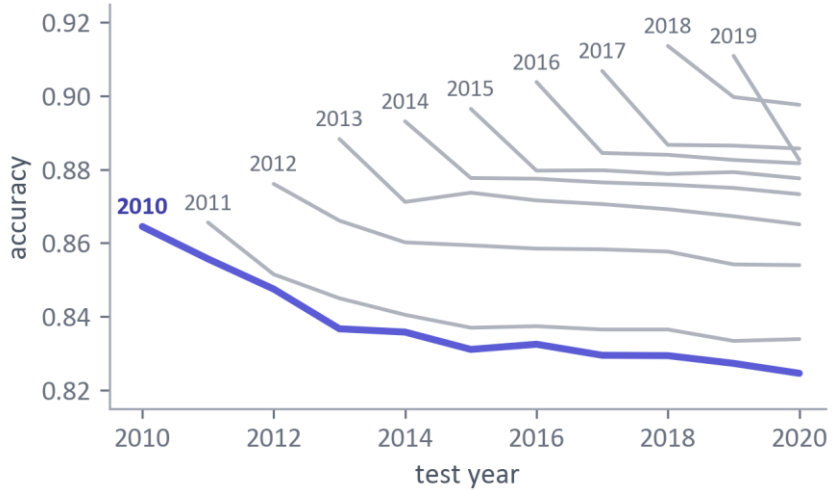
Entity ambiguity



Two entity linkers agree on only about 40% of entities

Understanding EL · EKAW 2024

Temporal drift



Every model scores lower on the years after it was trained

Graph-TempCZ · LREC 2026

Add the missing signal

Graph and its changes

TIGER · ECAI 2024
CYCLE · CIKM 2024



TIGER uses the links, CYCLE their changes.

TIGER +21%, CYCLE +18%,
over the 2023 state of the art, 3 years on.

Image semantics

Beyond Images · ESWC 2026





A vision-language model describes every image.

+5-8% Hits@1 overall,
7.5 → 32.5 on 20 logos.

Time

Time Imprint · under review



1769-1821



2020-2024

Each entity's years become an input of their own.

+6% Hits@1 overall,
+58% on the hardest 1% of look-alikes.

Recommendation and LLMs

Time-aware routing

TimeRoute · under review



Most days: text



Valentine's: image

Timestamps set each user's modality weights.

+5-10% over the strongest baseline,
on three datasets, ten seeds.

LLM-based selection

Select, Don't Train · ISWC 2026

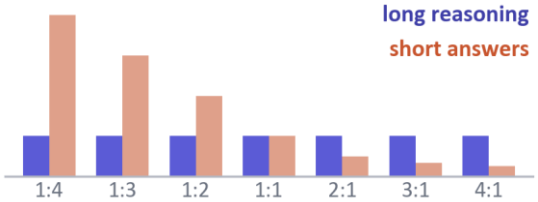


An LLM picks one of 16 candidates, or none.

86.3 vs 82.3 micro-F1 on ZELDA,
with no retriever training.

Thinking-mode data mix

Fusion Training · ACL 2026 SRW



Fine-tuning on long reasoning and short answers.

More short answers, weaker reasoning,
r = -0.58; interleaving is most robust.

Takeaways

- **Extra signal helps most where the model was weakest**
  - Each task has its own weak spot: entities with few links, logos, look-alikes, users with unusual habits.
- **The signal is often already in the data**
  - Timestamps, old graph versions, discarded images. No new labels, sometimes no change to the model.
- **An average hides both the problem and the gain**
  - Check later years and the hardest slice, not just the overall number.

What's next

Open questions

- **Is one more signal worth it?**
  - Predict which signal helps which entities, before training.
- **Are all link changes equal?**
  - Treat added and removed links apart, beyond yearly snapshots.
- **What breaks outside Wikidata?**
  - Product catalogues and domain knowledge graphs.

Where it applies

- **Search and recommendation**
  - Models drift between retrains.
- **Catalogue and entity matching**
  - Look-alikes, rare items, deduplication.
- **Knowledge bases behind RAG**
  - They go stale too.

One thread through 14 papers

Problem

Find where models fail: entity ambiguity and temporal drift.

PhD

Graph-TempCZ
Understanding EL
Survey (ESWA)

Solution

Add the signal models miss: the graph, how it changes, image semantics, time.

PhD

TIGER
CYCLE
Beyond Images
Time Imprint

MASTER

MVMA-GCN
RUNCL
Visual Analysis
DualChannel

Further

Use the same idea elsewhere: recommendation and LLMs.

PhD

TimeRoute
Select, Don't Train
Fusion Training

Pengyu Zhang

PhD candidate in Computer Science, University of Amsterdam · INDElab
p.y.zhang3@gmail.com · pengyu-zhang.github.io

Graph-TempCZ: A Graph Representation of Software Mentions for Predicting Software Usage in Scientific Publications

Congfeng Cao, **Pengyu Zhang**, Jelke Bloem · LREC 2026

Accuracy											
Train \ Test	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
2010	0.8645	0.8557	0.8475	0.8367	0.8358	0.8311	0.8325	0.8295	0.8294	0.8273	0.8246
2011	0.8687	0.8656	0.8515	0.8450	0.8405	0.8370	0.8374	0.8365	0.8365	0.8334	0.8339
2012	0.8799	0.8742	0.8761	0.8661	0.8602	0.8594	0.8585	0.8583	0.8577	0.8542	0.8540
2013	0.8933	0.8925	0.8860	0.8883	0.8712	0.8737	0.8716	0.8706	0.8692	0.8673	0.8651
2014	0.8906	0.8941	0.8878	0.8896	0.8931	0.8777	0.8775	0.8765	0.8759	0.8750	0.8733
2015	0.8914	0.8913	0.8901	0.8890	0.8890	0.8965	0.8797	0.8798	0.8788	0.8793	0.8776
2016	0.8849	0.8882	0.8876	0.8900	0.8912	0.8929	0.9038	0.8845	0.8840	0.8826	0.8817
2017	0.8885	0.8869	0.8899	0.8917	0.8941	0.8944	0.8945	0.9068	0.8867	0.8865	0.8857
2018	0.8989	0.8981	0.8997	0.9016	0.9022	0.9030	0.9051	0.9076	0.9136	0.8997	0.8976
2019	0.8861	0.8877	0.8868	0.8895	0.8893	0.8917	0.8934	0.8952	0.8945	0.9109	0.8826
2020	0.8888	0.8907	0.8955	0.8960	0.8975	0.8992	0.9018	0.9045	0.9061	0.9068	0.9200

93%

accuracy for GraphSAGE, about 6 points above the best feature-based baseline

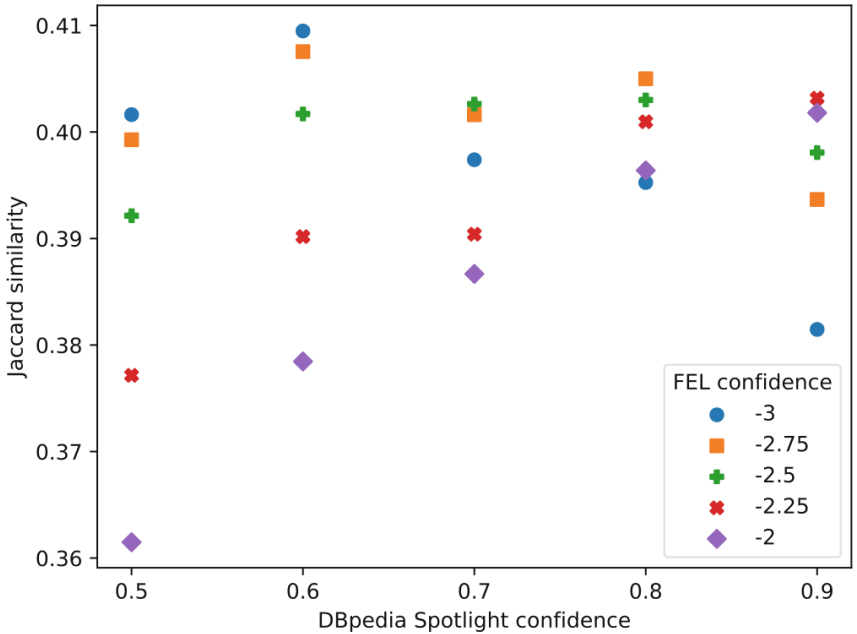
Problem Software usage had no graph and no time axis.

What we did A graph of 6.5M software mentions in biomedical papers; usage prediction as link prediction, one model per year, 2010-2020.

Result Every model scores lower on later years; transfer works for about two years. My part: the paper's logic, figures and revisions.

Understanding the Impact of Entity Linking on the Topology of Entity Co-occurrence Networks for Social Media Analysis

James Nevin, **Pengyu Zhang**, Dimitar Dimitrov, Michael Lees, Paul Groth, Stefan Dietze · EKAW 2024



~40%

overlap between the entities two linkers find in the same 21M tweets (Jaccard 0.38-0.41)

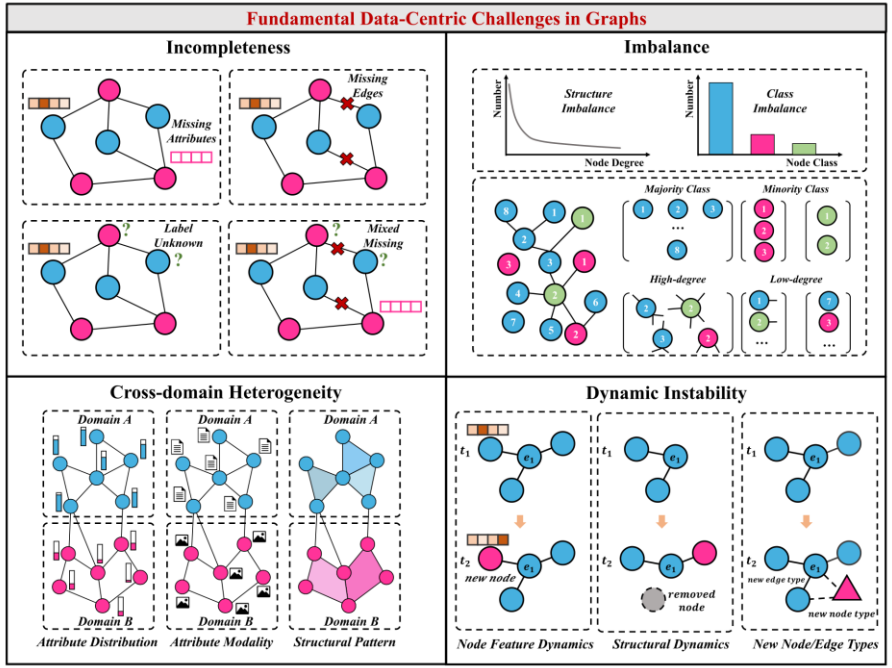
Problem With no ground truth for tweets, you can't pick the best linker, yet the network you build depends on it.

What we did Linked 21M COVID-era tweets with FEL and DBpedia Spotlight, five thresholds each, and compared the co-occurrence networks.

Result Size, connectivity and top entities shift with tool and threshold; COVID-19 can even leave the top ten. My part: the entity linking pipeline.

A Survey of Large Language Models for Data Challenges in Graphs

Mengran Li, **Pengyu Zhang**, Wenbin Xing, et al. · ESWA 2026



380+

papers kept from more than a thousand screened, grouped by four data challenges

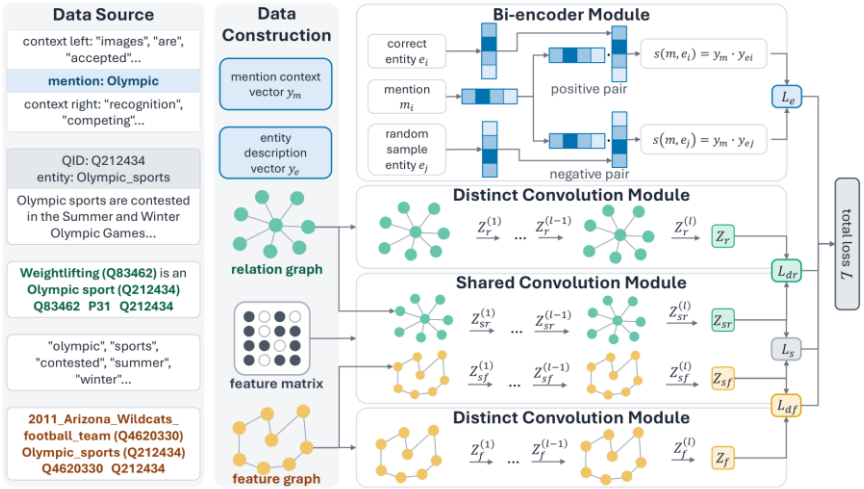
Problem Real graph data is incomplete, imbalanced, mixed across domains and keeps changing; LLM work on graphs lacked a map by these problems.

What we did Grouped papers by data challenge, with traditional and LLM methods side by side; an open repository goes with it.

Result A reading map and a list of open problems. My part: the dynamic instability and temporal sections.

TIGER: Temporally Improved Graph Entity Linker

Pengyu Zhang, Congfeng Cao, Paul Groth · ECAI 2024



+21%

over SpEL on continual entities, three years after training; 16% after one year

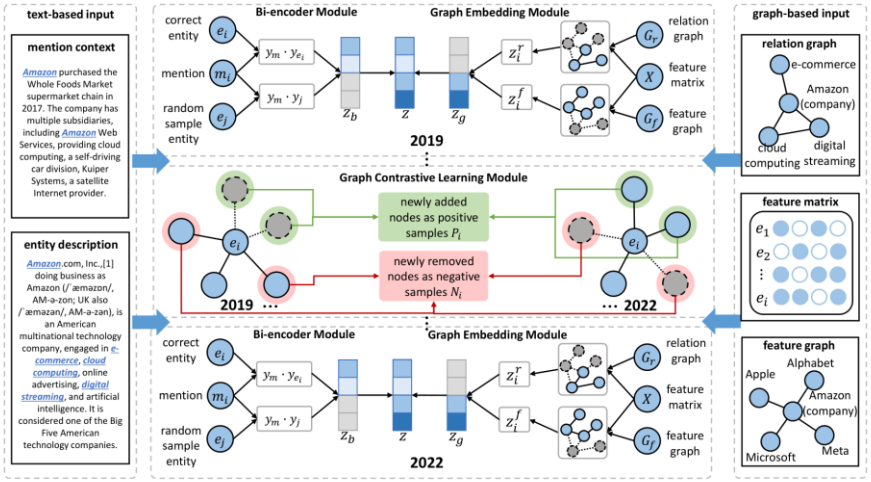
Problem Text-only entity linkers get worse as the knowledge graph changes.

What I did Added two graphs, Wikidata links and description similarity, with shared and separate GCN layers and an HSIC loss. The graph is used in training only.

Result 17% ahead with no gap, 21% at three years; entities with more links gain more. Data: 136k entities in four yearly snapshots (Graph-TempEL).

CYCLE: Cross-Year Contrastive Learning in Entity-Linking

Pengyu Zhang, Congfeng Cao, Klim Zaporojets, Paul Groth · CIKM 2024



+18%

over SpEL on continual and new entities, three years after training; 14% after one year

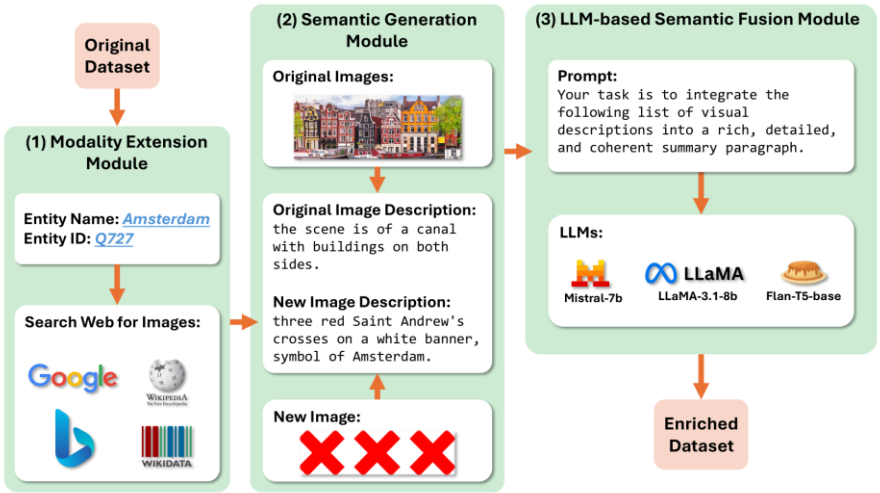
Problem TIGER treats each year's graph as fixed, so how the graph changes goes unused.

What I did Links that appear between snapshots become positives, links that disappear become negatives, with contrastive learning across the relation and feature graphs.

Result The biggest gains are on low-degree entities, the ones drift hurts most. Data: GCL-TempEL.

Are a Thousand Words Better Than a Single Picture?
Beyond Images - A Framework for Multi-Modal Knowledge Graph Dataset Enrichment

Pengyu Zhang, Klim Zaporojets, Jie Liu, Jia-Hong Huang, Paul Groth · ESWC 2026



7.5 → 32.5

Hits@1 on 20 logo and symbol entities; +5-8% overall across four models

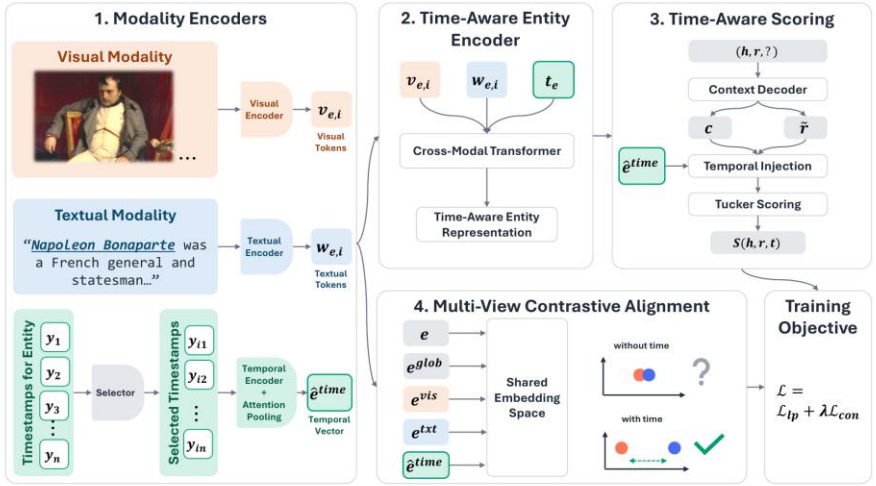
Problem Curated multi-modal KGs drop ambiguous images like logos, because their visual embeddings are noise.

What I did Collect every Wikipedia image of an entity, describe each with a vision-language model, and fuse the descriptions with an LLM into one summary used as text.

Result All four downstream models, used as they are, improve on all three datasets. The enrichment runs once, offline, before training.

Time Imprint: Learning Time-Aware Representations in Multi-Modal Knowledge Graphs

Pengyu Zhang, Klim Zaporojets, Congfeng Cao, Jia-Hong Huang, Paul Groth · under review



+58%

Hits@1 on the top-1% most ambiguous entities; +6% overall on MKG-W

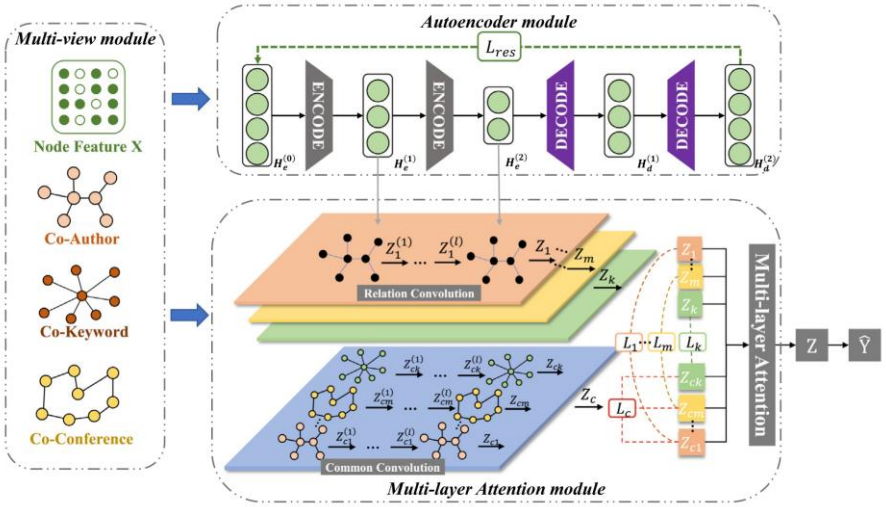
Problem Look-alikes share text and images; temporal KGs give time to triples, so entities have no time of their own.

What I did Pick a few years around the median plus the earliest one, pool them with attention, and inject them into the encoder, the scoring and a contrastive view.

Result New state of the art on MKG-W; with 40% of timestamps corrupted, Hits@1 drops less than a point.

MVMA-GCN: Multi-View Multi-Layer Attention Graph Convolutional Networks

Pengyu Zhang, Yong Zhang, Jingcheng Wang, Baocai Yin · EAAI 2023



92.7 vs 90.4

accuracy on ACM against AM-GCN, the strongest baseline (20 labels per class)

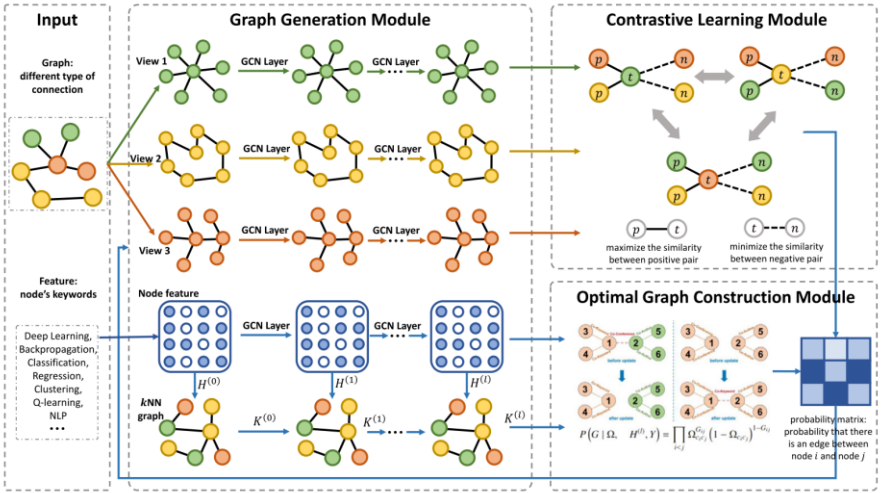
Problem Networks have several kinds of links; single-view GNNs drop them and can't weigh views or neighbours.

What I did All relation views plus a kNN feature graph, HSIC to keep each view distinct, attention over neighbours and over views, and an autoencoder branch.

Result Best in most settings on eight datasets; every module helps. TIGER reuses this design and the HSIC loss.

RUNCL: Relationship Updating Network with Contrastive Learning

Pengyu Zhang, Yong Zhang, Xinglin Piao, Yongliang Sun, Baocai Yin · Physica A 2024



3% → 6%

gain over the strongest baseline on low-degree nodes as labels drop from 20 to 5 per class

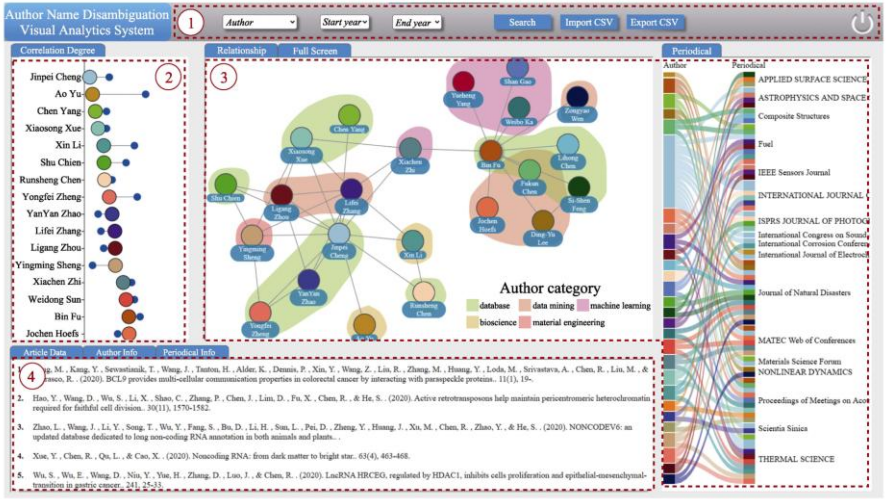
Problem Missing or wrong edges hurt poorly connected nodes most, but GNNs trust the input graph.

What I did Candidate kNN graphs from node features, the best one picked with a stochastic block model, then contrastive learning between feature and relation views.

Result Best in most low-label settings on six benchmarks. CYCLE reuses the low-degree focus.

Visual Analysis for Name Disambiguation of Academic Papers

Pengyu Zhang, Yong Zhang, Yanjie Cui, Baocai Yin · JCAD 2022, in Chinese



95%+

task success for eleven volunteers on 4,000 real university papers; usability 84-88 out of 100

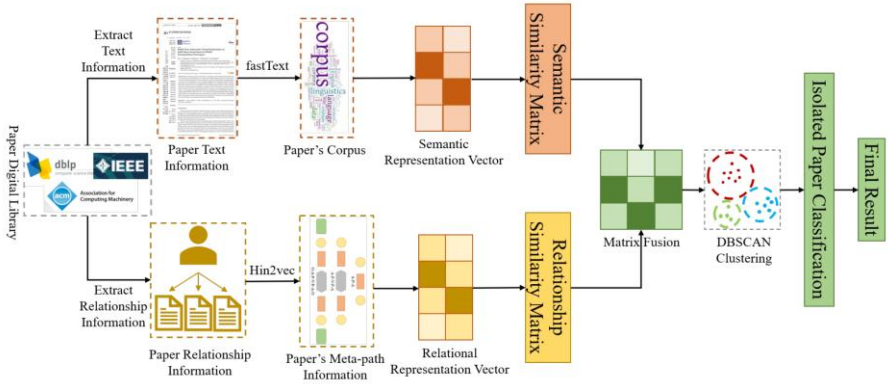
Problem Romanised Chinese names collide, and existing tools were hard for non-experts to use or check.

What I did A multi-view GCN predicts each author's research field, and linked views show collaborations, journals and raw metadata.

Result About 91% accuracy on ACM and DBLP; two real cases confirmed by research administrators.

Dual-Channel Heterogeneous Graph Network for Author Name Disambiguation

Xin Zheng, **Pengyu Zhang**, Yanjie Cui, Rong Du, Yong Zhang · Information 2021



F1 0.62

on AMiner WhoIsWho, above six baselines, with balanced precision and recall

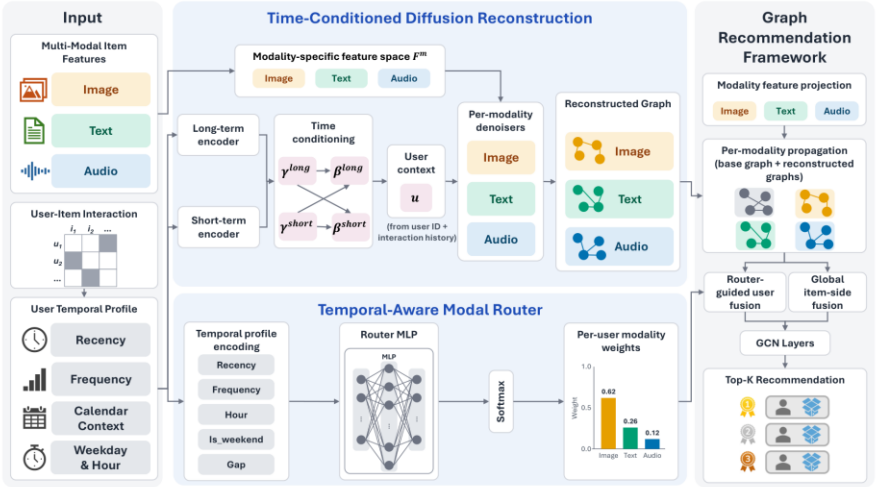
Problem Name disambiguation methods used either text or graph structure, rarely both.

What we did A text channel (fastText) and a relation channel (co-author, organisation and venue graph with Hin2Vec), similarities added, then DBSCAN clustering.

Result 0.62 against 0.61 for the best baseline; best for names with fewer homonyms and enough papers. My part: the manuscript, with all figures and tables.

TimeRoute: Time-Aware Modality Routing and Diffusion for Multi-Modal Recommendation

Pengyu Zhang, Yangqin Jiang, Klim Zaporojets, Congfeng Cao, Paul Groth · under review



5-10%

over the strongest baseline on TikTok and two Amazon datasets, ten seeds

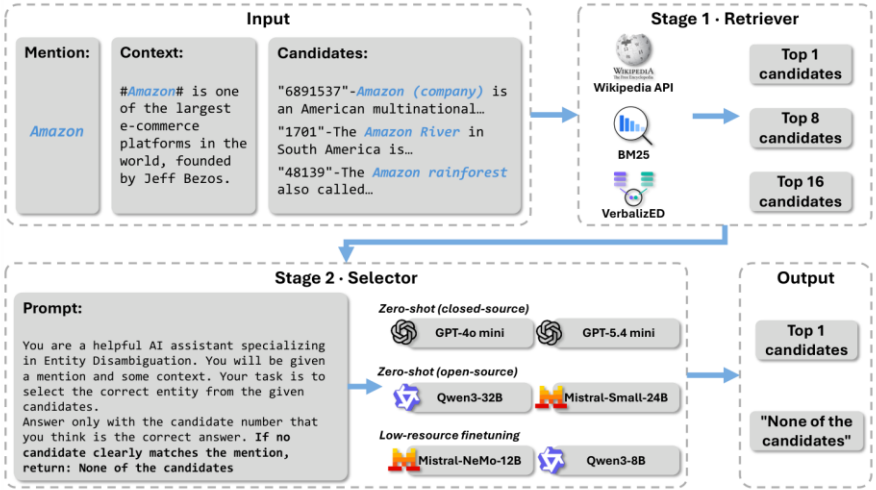
Problem Which modality matters drifts over time and differs by user; one global weight can't follow that.

What I did A router turns each user's timestamp profile into modality weights, with under 700 parameters; a time-conditioned diffusion module filters outdated edges.

Result Random noise instead of timestamps removes the gain (Amazon-Baby); the routing shifts with user activity.

Select, Don't Train: The Benefits of Modular Entity Disambiguation with LLM-Based Selection

Fina Polat, Daniel Daza, **Pengyu Zhang**, Klim Zaporojets, Paul Groth · ISWC 2026



86.3

micro-F1 on ZELDA with untrained BM25 and an LLM selector, 4 points above the best dual encoder

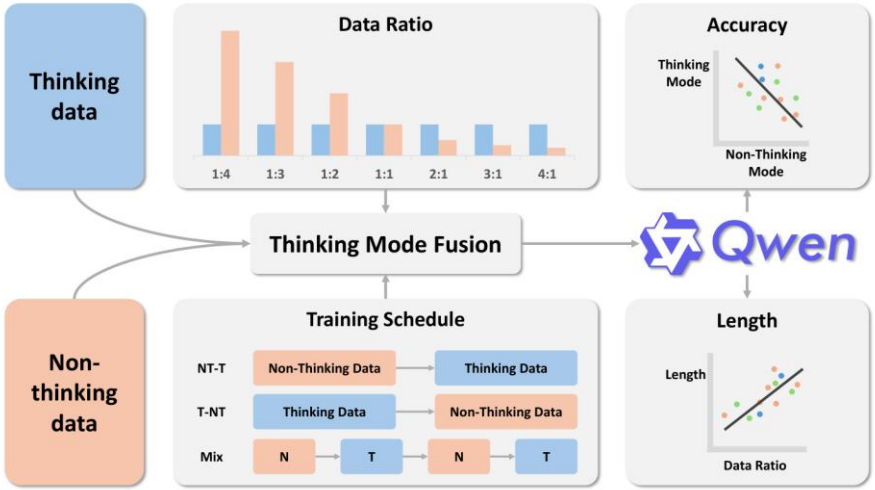
Problem Dual encoders squeeze retrieval and selection into one space and need retraining as the KB changes.

What we did Split the two: a retriever returns 16 candidates, an LLM picks one or abstains. Retrievers: Wikipedia API, BM25, or a trained dense model.

Result A trained dense retriever lifts it to 88.5, and 90.7 when correct abstentions count. My part: the discussions throughout; Fina led the work.

Fusion Training for Mathematical Generalization in Large Language Models

Congfeng Cao, **Pengyu Zhang**, Jelke Bloem · ACL 2026 SRW



r = -0.58

correlation between thinking and non-thinking accuracy across ratios and schedules

Problem Hybrid models learn long reasoning and short answers together; the right mix was unknown.

What we did Fusion Bench: 1,500 thinking examples mixed with 375 to 6,000 non-thinking ones (seven ratios) in three schedules; Qwen3-4B with LoRA, three runs each.

Result More short answers lower reasoning accuracy (0.224 to 0.199); interleaving is most robust. My part: the paper's logic, figures and revisions.

Thesis overview

Addressing Temporal Dynamics and Entity Ambiguity in Knowledge Graph Tasks with Multi-Source Information

Hypothesis: information sources not yet used as modeling signals improve KG tasks under temporal dynamics and entity ambiguity

	Challenge 1: Temporal dynamics (RQ1)		Challenge 2: Entity ambiguity (RQ2)	
	Chapter 2 · RQ1.1 TIGER + CYCLE	**Chapter 3 · RQ1.2** TimeRoute	**Chapter 4 · RQ2.1** Beyond Images	**Chapter 5 · RQ2.2** Time Imprint
Task	Entity Linking	Multi-modal recommendation	Link prediction in MMKGs	Link prediction in MMKGs
Problem	Text-only linkers degrade as the KG evolves	Modality relevance drifts per user; global fusion cannot track it	Ambiguous images (logos, symbols) act as noise and are discarded	Look-alike entities share text and images
Signal	Graph structure and its evolution	Users' temporal contexts	Text distilled from ambiguous images	Time as an entity-level modality
Approach	Graph convolution over snapshots + cross-year edge contrastive learning	Per-user modal router + time-conditioned diffusion reconstruction	Image retrieval + image-to-text conversion + LLM fusion	Three-view contrastive alignment + timestamp selection
Finding	**Complementary gains across high- and low-degree entities**	**Gains need both the mechanisms and temporal input**	**Largest gains on logo and symbol images**	**Largest gains on the most ambiguous entities**